

Aswini Leela Loganathan^{1,2}, Heerman Kumar¹, Harpreet Kaur Ranjit Singh¹, Qasim Ayub^{1,2}

¹School of Science, Monash University Malaysia, Jalan Lagoon Selatan, Bandar Sunway, 47500 Selangor Darul Ehsan, Malaysia

²Monash University Malaysia Genomics Platform, Monash University Malaysia, Jalan Lagoon Selatan, Bandar Sunway, 47500 Selangor Darul Ehsan, Malaysia

Email: aswini.leela@monash.edu

BACKGROUND AND OBJECTIVES

- Human genetic diversity is shaped by demographic history and natural selection. Indigenous populations from ecologically contrasting environments are powerful models for studying the evolution of immune genes driven by parasites and pathogens.
- Contrasting ecological settings:**
- SE Asia (Malaysia): tropical rainforest; high parasite diversity: *P. knowlesi*, helminths, dengue
- South Asia (Pakistan): arid & temperate; distinct pathogen exposure histories

The aims of this study are:

- To reconstruct the demographic histories of indigenous and isolated populations from Malaysia and Pakistan by inferring effective population size changes, divergence, and admixture.
- To detect signatures of parasite and pathogen-mediated selection at innate and adaptive immunity loci across contrasting ecological and epidemiological environments.
- To assess the relative contributions of demographic history and infectious disease pressure to present-day immune gene diversity by integrating neutral genomic baselines with targeted selection analyses.

Study Population

Table 1: Study Population: Orang Asli ("original people") are among the world's most understudied populations almost entirely absent from gnomAD, 1000 Genomes and HGDP.

Population	n	Location and Lifestyle	Status
Jahai	30	Negrito hunter-gatherers, North Malaysia	Sequenced
Temuan	40	Senoi agro-foragers, South Malaysia	Sequenced
Parsi	TBC	Endogamous community, Pakistan	In progress
Pakistani tribal	TBC	Highland/arid zone, Pakistan	In progress

METHODOLOGY

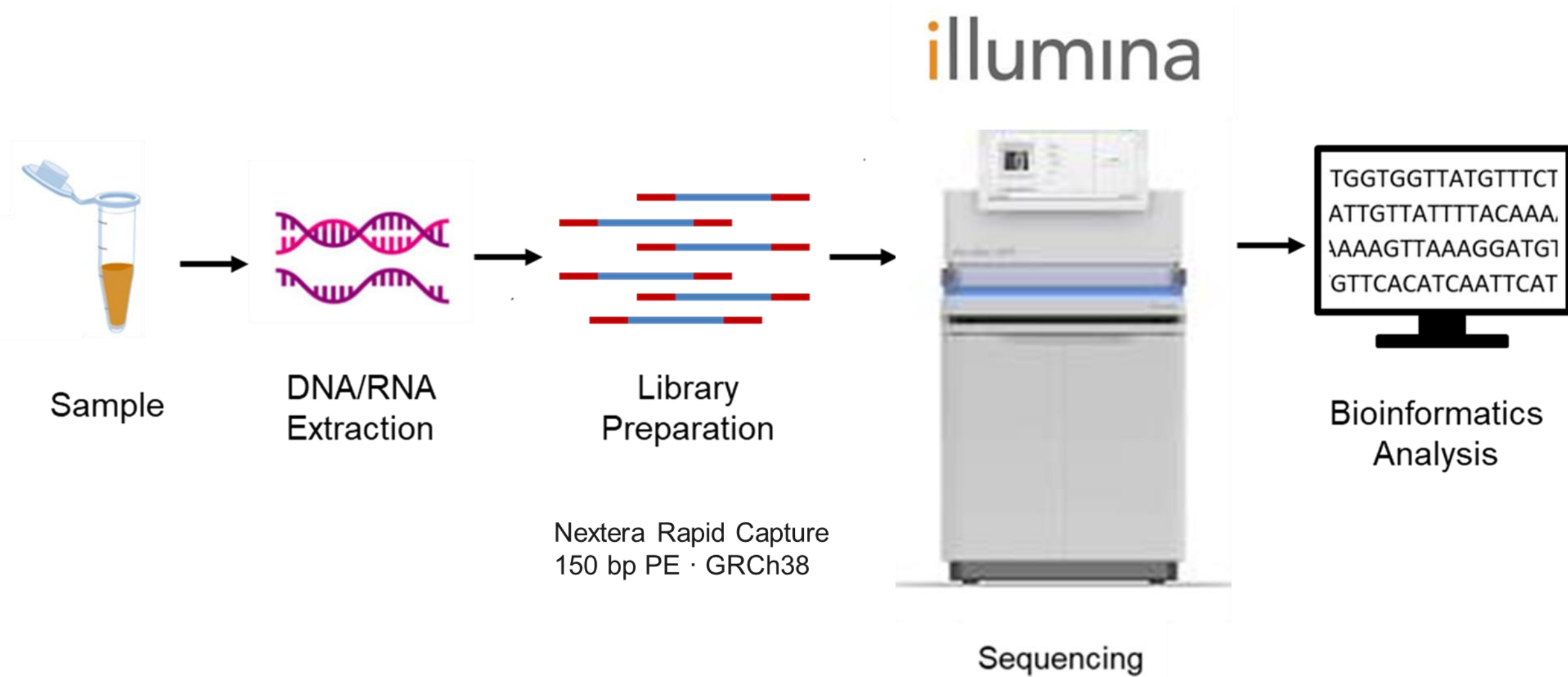


FIGURE 1: Wet-lab workflow: Genomic DNA was extracted using an organic method, ligation libraries were prepared, and sequenced on the Illumina NovaSeq 6000 to generate the reads.

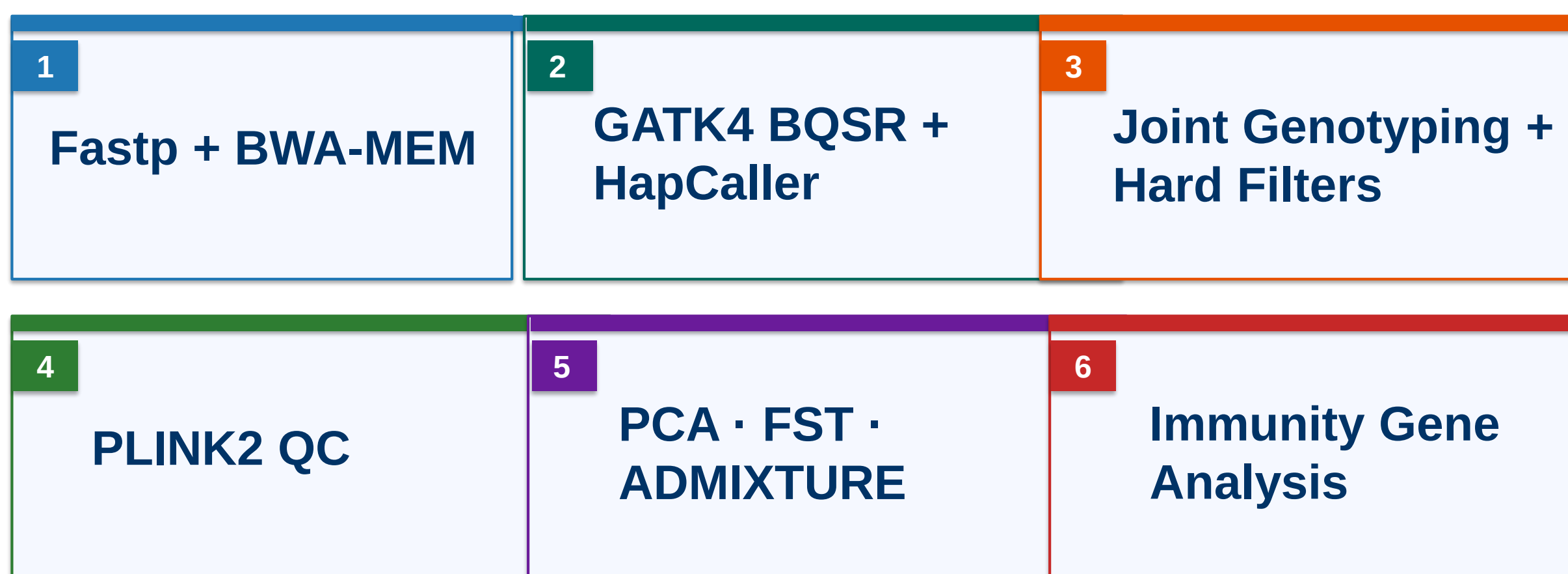


FIGURE 2: Bioinformatics workflow used for this study.

SEQUENCING STATISTICS

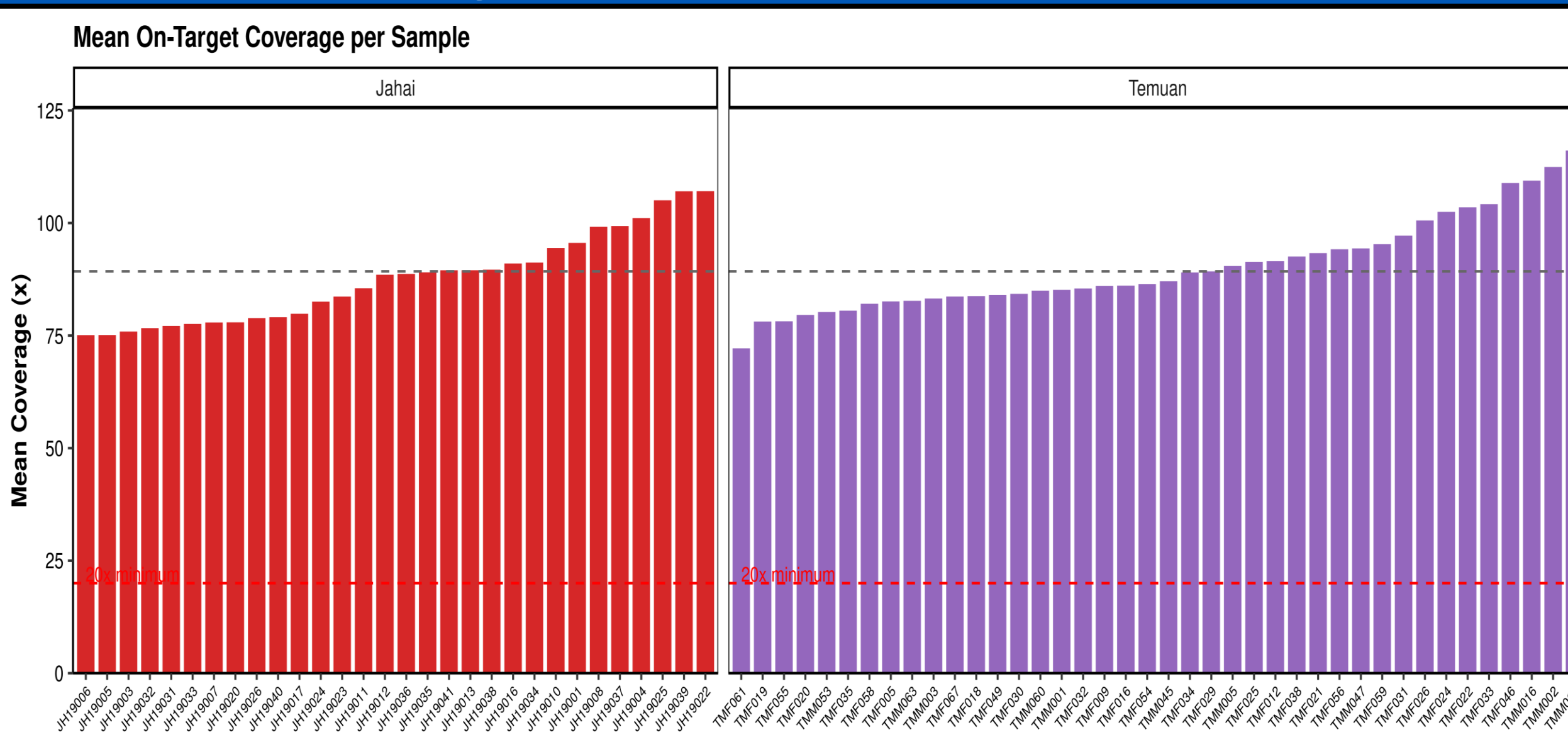


FIGURE 3: Sequencing Depth per Sample (WES, n=70)

RESULTS

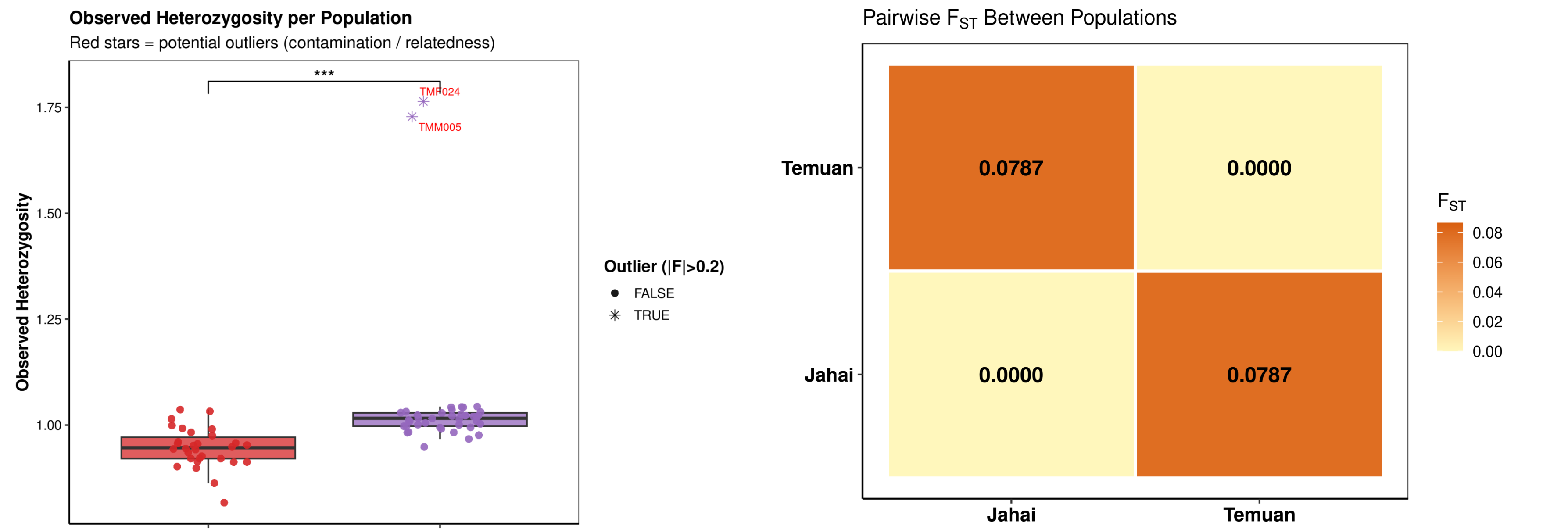


FIGURE 4: (A) Individual Heterozygosity (Inbreeding Coefficient F). (B) Pairwise Population Differentiation (F_{ST}) Heatmap. $F_{ST}=0.079$ reflects deep-time divergence between Jahai and Temuan.

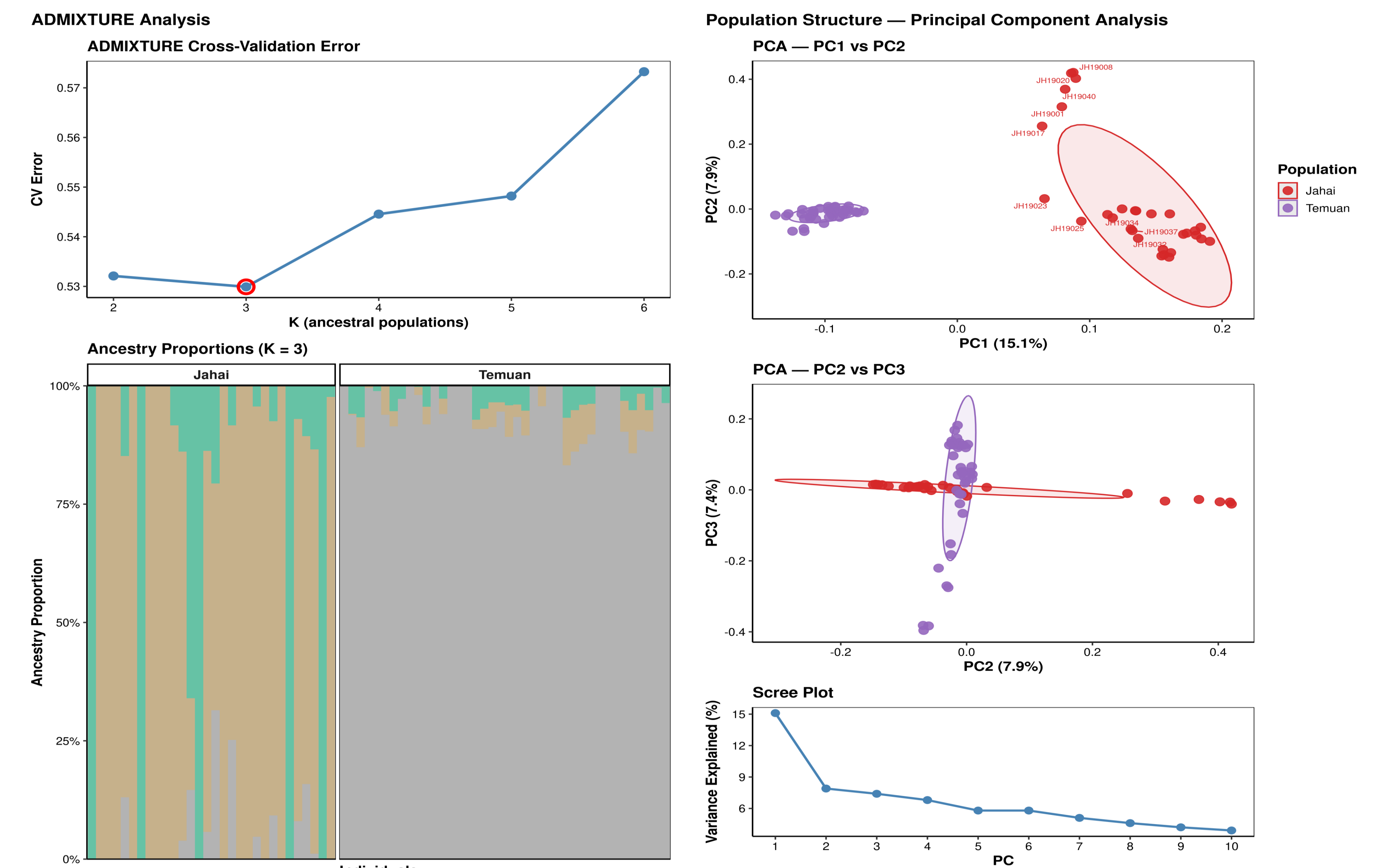


Figure 5: (A) ADMIXTURE Ancestry Proportions ($K = 2-6$). (B) Principal Component Analysis ($PC1=15.1\%$ · $PC2=7.9\%$ · $PC3=7.4\%$). Complete PC1 separation confirms deep-time divergence; Temuan carry a Malay/Austronesian component absent in Jahai, consistent with a later Austroasiatic migration.

Immunity and Parasite-Resistance Gene Landscape

Whole-exome sequencing variants in 23 innate/adaptive immunity and malaria-resistance loci

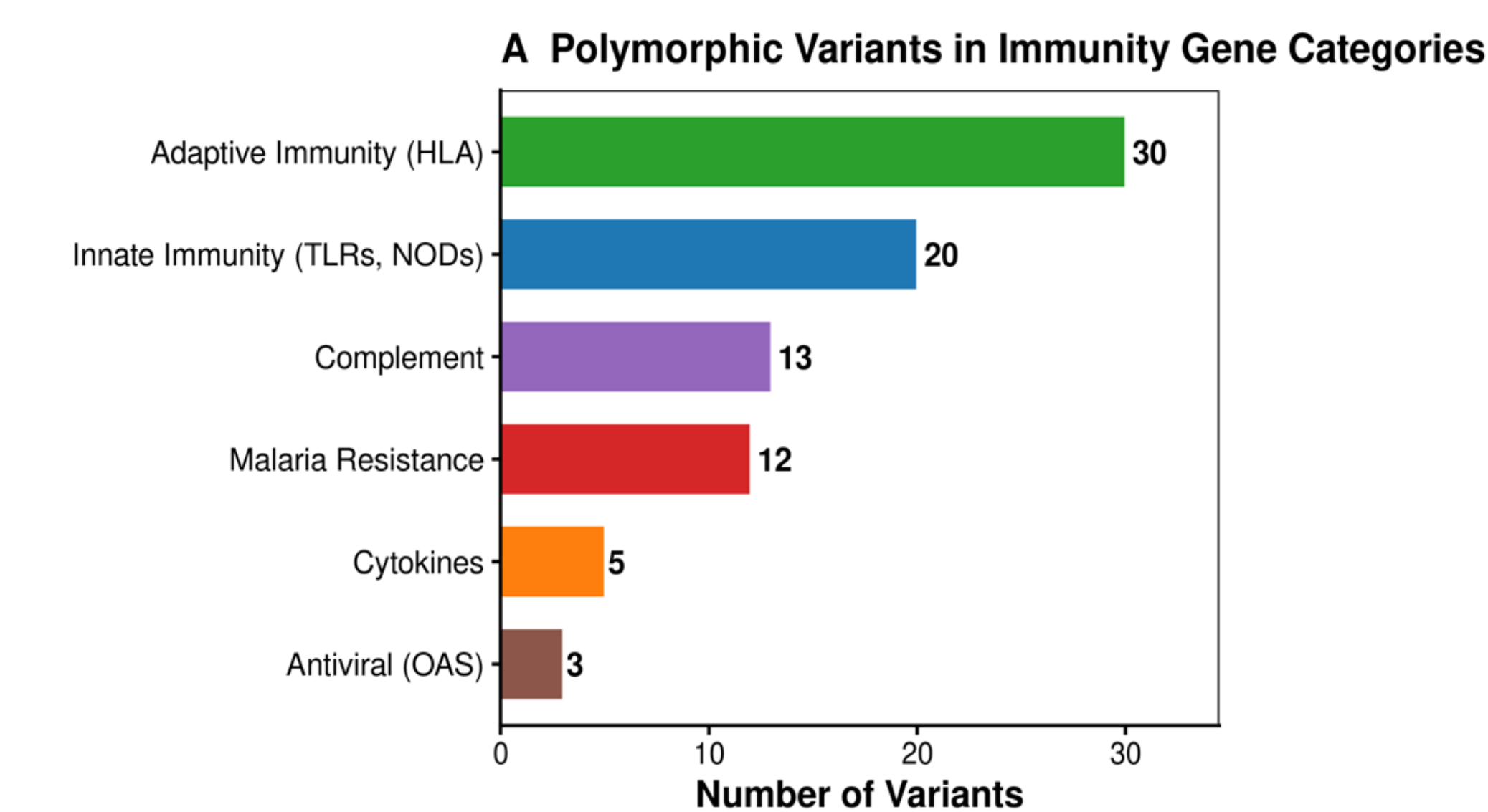
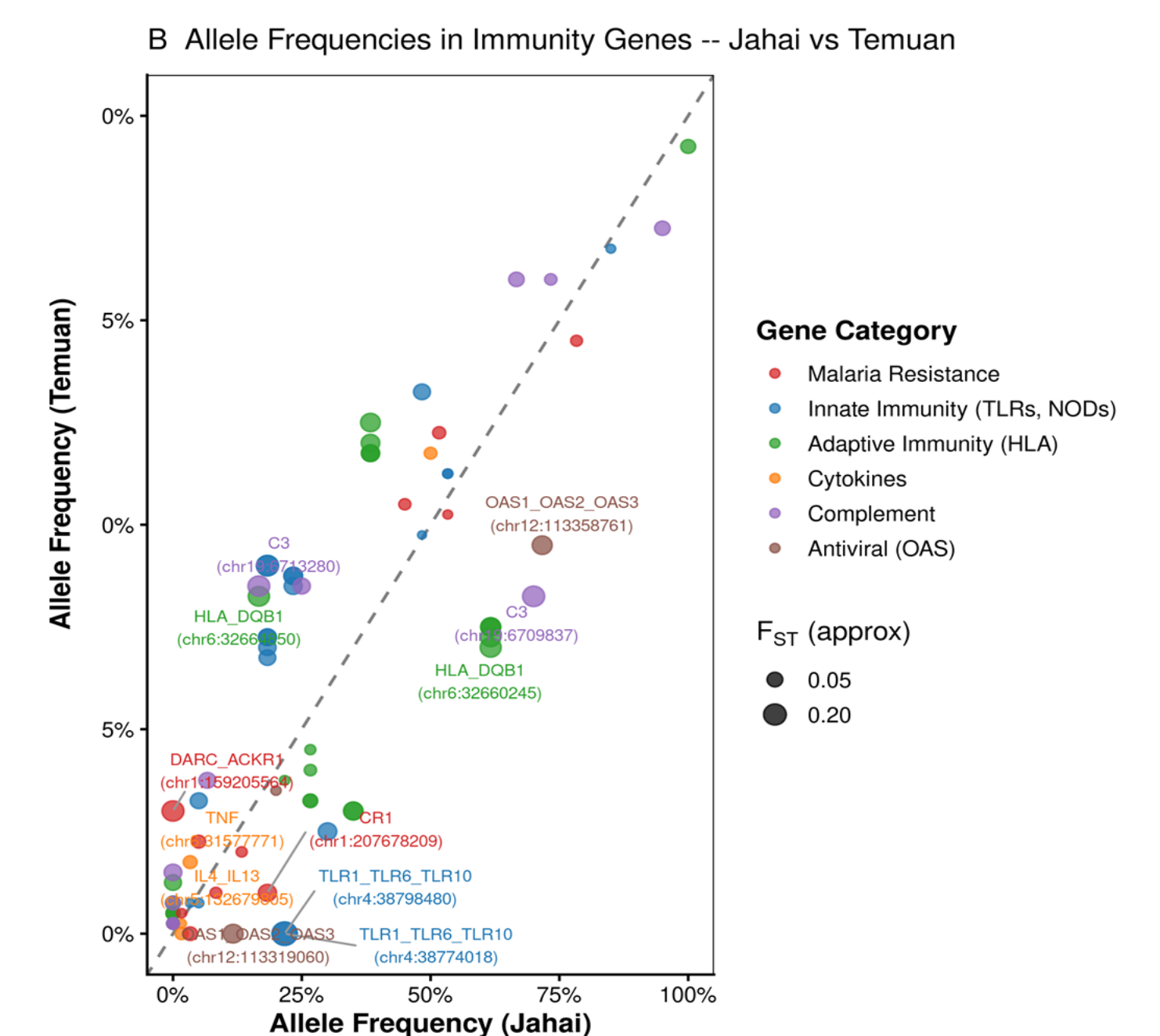


Figure 6: (A) Immunity & Parasite-Resistance Gene Landscape (23 loci, 83 variants). Elevated F_{ST} at TLR1/TLR6/TLR10 and DARC/ACKR1 loci indicates population-specific immune adaptation, reflecting divergent helminth and malaria exposure histories.



DISCUSSIONS & FUTURE DIRECTIONS

Population Structure

- Complete PC1 separation (15.1% variance) — zero overlap between Jahai and Temuan. Strong, biologically real genetic differentiation.
- $F_{ST} = 0.079$ (Hudson) — distinct deep-time origins: Jahai from first human dispersal into SE Asia (~50–60 kya); Temuan from a later Austroasiatic migration.
- $K = 3$ best ADMIXTURE fit — Temuan carry a Malay/Austronesian component entirely absent in Jahai.

Inbreeding & Diversity

- Jahai mean $F = +0.054$ — mild excess homozygosity; small (~3,000), forest-isolated hunter-gatherer group.
- Temuan mean $F = -0.047$ — excess heterozygosity; admixed history. TMM024 & TMM005 flagged for kinship check.

Immunity Gene Landscape

- TLR1/TLR6/TLR10 — $F_{ST} \approx 0.28$: allele at 21.7% Jahai, absent Temuan. Helminth & mycobacterial susceptibility.
- DARC/ACKR1 — *P. vivax* resistance variant absent in Jahai, 15% in Temuan. Differential malaria exposure histories.
- HLA-DQB1 — $F_{ST} \approx 0.14-0.16$; divergent pathogen-driven adaptive selection.

Together, these results show that demographic isolation and parasite-driven selection have jointly shaped immune gene diversity, with each population carrying a distinct adaptive legacy.

Extend demographic inference and selection analyses to the Pakistan cohort (Parsi and Pakistani tribal) to enable cross-continental comparison of immune gene evolution.

Apply simulation-informed machine-learning to genome-wide selection scans, improving power to distinguish adaptive from demographic signals in small populations.

Reconstruct genome-wide genealogies using coalescent approaches to resolve fine-scale population history and admixture timing across all four cohorts.